

EV Asset Performance Management Agent — Detailed Document

ET AI Hackathon 2026 · Team: V. Abhishek Prakash (Layer 1 + integration), Akhil Prasad (Layer 2 + data audit), Hrishikesh (testing + documentation)

Every quantitative claim below traces to `docs/data_audit_final.md` (numbered audit flags) or a committed notebook; this document does not re-derive them.

1. Problem & context

EV charge-point operators run critical infrastructure with no operational rigor: chargers fail, and the operator finds out when a customer complains. Downtime means lost revenue and stranded drivers, and technicians are dispatched blind — often to faults that had already cleared themselves.

The telemetry that would have caught these faults already exists. Every OCPP-compliant charging-management system (CMS) streams status notifications, meter values, and session lifecycle events. Nobody is reading it for fault signatures.

What we built is **AI Maintenance Decision Support for EV charging infrastructure**: an agent that watches the CMS event stream and turns raw telemetry into prioritized, explainable maintenance recommendations — each carrying a maintenance priority, confidence, likely root cause, recommended action, and impact class. It tracks charger *behavior* over time — a behavior proxy, explicitly **not** battery state-of-health or state-of-charge diagnostics. It runs beside any OCPP-compliant CMS: replaying historical exports today, consuming a live stream in production.

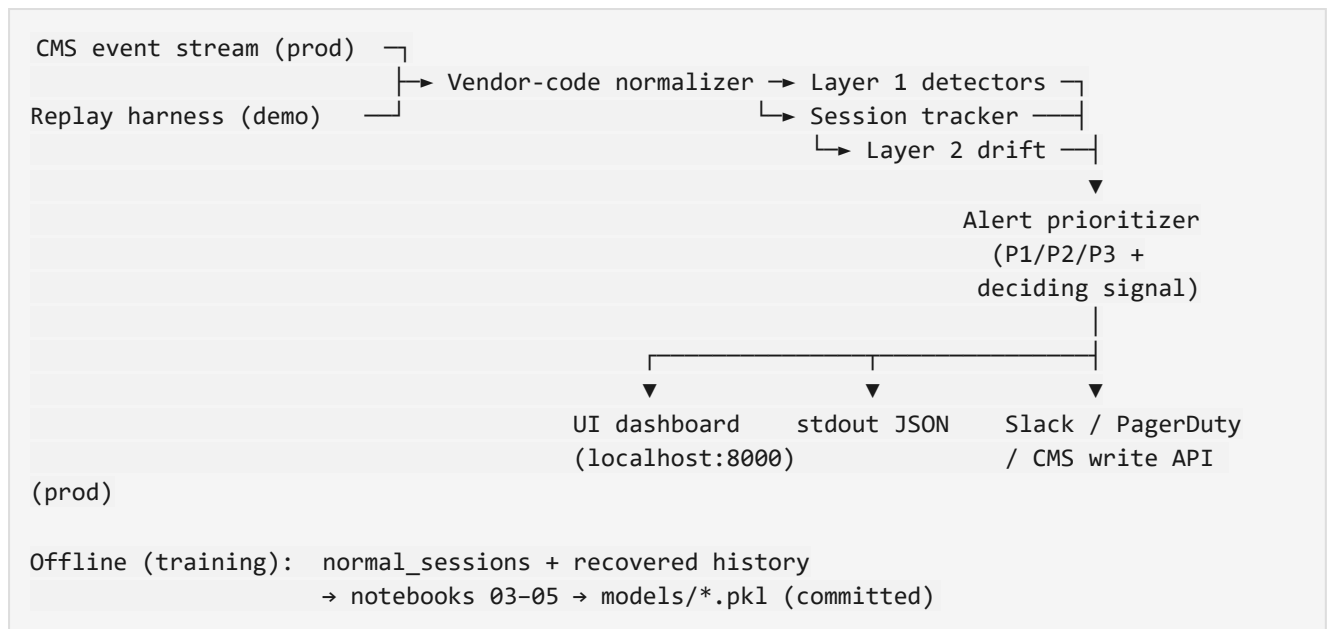
2. Approach and two-layer architecture

Two detection layers with complementary failure modes:

- **Layer 1 — deterministic.** State machines and point-event detectors for known fault signatures, fed through a vendor-code normalizer that resolves the delivered taxonomy's 659 canonical vendor codes onto the 19 OCPP-standard categories (the source CMS logs 20,202 distinct vendor strings — `data/reference/source_vendor_code_counts.csv`). Deterministic detection is precise, explainable, and needs no training data — the right tool for faults whose signatures are documented.
- **Layer 2 — unsupervised drift.** A per-connector Isolation Forest scores every closed charging session against that connector's own learned baseline, catching degradation that has no named fault code. Sparse connectors fall back to vendor-family pools, then a global pool.

Between detection and every alert sink sits an **alert prioritizer** that assigns each alert a P1/P2/P3 tier and a human-readable deciding signal. This is the alert-fatigue answer and the project's business

centerpiece: a fault that self-recovers in seconds is not the same event as one that never recovers, and the operator should never see them at the same priority.



Full diagram: `docs/architecture_v2.svg` (Mermaid source `architecture_v2.mmd`).

Three containerized services — `replay`, `detector`, `ui` — brought up by a single `docker compose up`. Replay emits time-ordered JSON events; the detector consumes them, runs both layers, and streams prioritized alerts; the UI renders a live operator dashboard. No cloud dependency.

3. Data — source scale vs delivered/validated

Every scale figure in this document carries one of two tags, matching `docs/claims_evidence.md` exactly: **[SOURCE-SCALE]** — the size of the originating production CMS, evidenced by aggregate-count CSVs under `data/reference/` but *not* reproducible from the delivered slice — or **[VALIDATED]** — reproducible from the committed repo right now, with a reproduction command in `claims_evidence.md`. A source-scale figure is never presented as a validated one.

Anonymized OCPP telemetry exports from a production CMS. Governance is enforced in-repo: raw CMS data is never committed (gitignored), charge-box IDs ship as SHA-256 hashes, geo coordinates are rounded to ~1 km, and customer/RFID/IP fields are dropped at export. A committed audit script (`tests/anonymization_audit.sh`) gates the repo and **caught real PII in a delivered taxonomy before commit** — 16,260 card-tag values, 3,807 subscriber-phone rows, and 950 routable IPs, all masked (audit flag 22).

Two-tier fleet framing. [SOURCE-SCALE]: 655 chargers across 130+ manufacturer families and 209 manufacturer-model configurations, 33.5M events (3.4M status + 30.1M telemetry), and 103,081 sessions (`manufacturer_inventory.csv`, `source_event_totals.csv`, `source_session_count.csv`). **[VALIDATED]** — what the system was built and measured on: **39 stations, 80 connectors, 19 firmware versions, and 10,090 normal sessions** (`total_stations.csv`, `charger_stations.csv`, `normal_sessions.csv`).

Real fault data arrived late and in pieces, each analyzed and documented:

- `error_taxonomy.csv` — 17,954 rows, 19 OCPP categories, **659 distinct canonical vendor codes [VALIDATED]** (the delivered working taxonomy). The source CMS logs 20,202 distinct vendor strings over 1.74M occurrences **[SOURCE-SCALE]** (`source_vendor_code_counts.csv`); the delivered 659 resolve 100% (flag 22).
- `missing_real_faults.csv` — 79,681 real fault events: GroundFailure 79,480, WeakSignal 150, OverVoltage 51 (flag 22).
- `err1024_err1051_status_sequences.csv` — 21,910 rows of real err-code sequences across 6 connectors (flag 23).
- `err1024_crash_signature.csv` — the err1024 mechanism, 3 real events (flag 21).

Data-driven limitations are stated plainly in §8 — they are part of the evidence trail, not a footnote.

4. Detection methods — six categories, deterministic

Layer 1. `err1051` is a five-step state machine (IDLE → first-seen → meter-zero → second-seen → stop → recovery) that classifies transient (≤ 15 s) versus technician-dispatch (> 15 s) recoveries. `err1024` is a point-event handler confirmed final by its crash signature — "energized but never charging" (227 V nominal, current/power ≈ 0), no retry sequence to model (flag 21). Telemetry-silence fires when an active session goes quiet past a threshold. WeakSignal, GroundFailure, and Under/OverVoltage are point-event detectors keyed on the canonical category from the vendor-code normalizer. A schema difference between fixtures and real streams — system-err codes live in `vendor_error_code` with `error_code=OtherError` — was found on the real export and handled by matching either column (flag 23).

Prioritizer. Rules with tunable weights at the top of the file: transient `err1051` → P3, unrecovered → P1, GroundFailure → P1 (safety), and data-driven escalations — WeakSignal bursts, 24-hour voltage repeats, ≥ 3 consecutive drift sessions, repeat offenders, and a **station-wide** rule (a second connector of the same station faulting within 60 s → P1), motivated by the finding that **88% of real fault episodes hit both plugs within 5 s** (flag 19).

5. Layer 2 degradation tracking — and the honest lead-time negative

Layer 2. Per-connector Isolation Forests (`contamination='auto'`, `random_state=42`) on a `duration_sec + start_hour` feature vector, with vendor-family and global pooled fallback for sparse connectors. The SPEC's default threshold -0.1 flagged $\sim 6\%$ of held-out normals; we **calibrated** to -0.1187 (notebooks 03–04). An expanded-normal retrain from reconstructed session history shipped only after passing an FPR-non-degradation gate (flag 17).

Evaluation. The headline metric is measured on a **chronological** per-connector 80/20 holdout (most recent 20% held out — production-mimicking), not a random split.

The lead-time negative, published rather than hidden. A lead-time analysis tested whether Layer 2 drift precedes Layer 1 faults; it does not (lift $0.57\times$ — `docs/layer2_leadtime.md`). Layer 2 is

therefore framed honestly as **orthogonal degradation tracking**, never as early warning or predictive maintenance (flag 17). We regard publishing this negative result as a feature of the submission.

6. Results

All results in this section are [VALIDATED]-tier — reproducible from the committed repo; reproduction commands in `docs/claims_evidence.md`.

- **False-positive rate: 3.62%** on the chronological held-out split ($n = 2,015$, at the calibrated threshold -0.1187) — comfortably under the 5% acceptance bar. Per tier: heavy 3.3% / medium 3.5% / light 3.2% / pooled 6.1%; `siemens` family 10.5% ($n=76$) on the watch list (flag 14). Stated for completeness: at the **untuned SPEC default threshold (-0.1) the rate is 5.26% — above the bar** (`docs/layer2_scope.md`), which is precisely why the threshold was calibrated (random-split calibration reference 4.02%, notebook 03). Both numbers are published; only the calibrated one is claimed against the bar.
- **Vendor-code coverage: 100% of the 659 delivered canonical codes resolve to a labeled OCPP category**, 0% unlabeled residue. Framed honestly: shape rules decide a minority (152 of 659) and the already-labeled `error_code` field the rest — "resolves to a labeled category", never "routed by rules" (flag 22).
- **Real-event detection: 100% per category** — `err1024` 99/99, GroundFailure 79,480/79,480, WeakSignal 150/150, OverVoltage 51/51, UnderVoltage 637/637 (plus 18/18 on the capped export). **All six categories are real-event verified:** `err1051`'s machine fired end-to-end on 88/88 real status-spine sequences and 2/2 episodes of a full connector-month with transaction linkage; only its meter-zero corroborating step remains fixture-only (flags 18, 22, 23, 25, 27). (The dashboard's coverage counter reads "7 of 19" on mixed streams — it counts Under- and OverVoltage as separate OCPP categories; this document counts voltage as one.)
- **The 13-second story, vindicated on real data:** across 190 real `err1051` events, median recovery is 10 s and **80% self-recover within 15 s** — so the P3 downgrade removes exactly that noise. Across *all* fault types only ~42% self-clear ($n=2,041$), which is why every other category defaults to P2/P1 (flags 12, 23).
- **Pipeline:** `docker compose up` from a fresh clone brings all services up with the UI answering HTTP 200; 98 automated tests pass (95 on a fresh clone — 3 skip without a gitignored real-data file); the anonymization audit passes.

7. Deployment

The production shape is a **sidecar container** beside the operator's existing CMS: it subscribes to the event stream (REST poll, message queue, or webhooks) and pushes prioritized alerts to Slack, PagerDuty, or the CMS's own write API. `ALERT_SINK` toggles between stdout (development) and HTTP (the UI or a production sink). There is no cloud dependency and no vendor lock-in — the same container runs on a laptop or beside a fleet's CMS. Model training is offline; only the committed `.pk1` artifacts are loaded for inference, so inference runs on any hardware.

Scale path: a CMS vendor licenses the detection logic and bundles it as a platform feature.

8. Limitations — stated plainly

Every limitation below is documented in a committed source; we consider this section a strength of the submission, not an apology.

- **No predictive lead time.** Layer 2 drift does not precede Layer 1 faults in our data (lift 0.57×, `docs/layer2_leadtime.md`). We claim degradation *tracking*, never prediction.
- **Thin Layer 2 feature vector.** `duration_sec` + `start_hour` only — the measurand-rich features (temperature, power shape) are implemented and tested but unvalidatable on the delivered exports (`docs/layer2_scope.md`).
- **Temperature is out of scope.** All-zero in every fault-window export (flags 11/20); later deliveries proved real values exist at source (28–66 °C, n=79, flag 26) and added 65,994 located readings (flag 27) — both after the scope freeze, so the features stay out and are ledgered.
- **err1051's meter-zero step is fixture-only.** The machine is real-verified end-to-end on its status spine (88/88, flag 25) and on a connector-month with transaction linkage (2/2, flag 27); no delivered fault window carries meter rows, so that one corroborating step has never fired on real data.
- **Telemetry-silence has no confirmed real positive.** Its 229 firings on the capped export were export-truncation artifacts — documented as such (flag 13). The detector is validated by unit tests and fixtures.
- **Untuned threshold sits above the bar.** 5.26% FPR at the SPEC default; the claimed 3.62% requires the calibrated threshold (−0.1187).
- **Streaming ingestion tails a file, not a live socket.** The `--follow` mode processes appended frames at arrival cadence; a live CMS WebSocket transport remains future work (`docs/future_work.md`).
- **Operational scope.** 6 of 19 OCPP categories have dedicated detectors; GroundFailure needs episode dedup before production (79,480 events from one chattering-sensor cohort); the dashboard's buffers are in-memory and reset on restart.

9. Future work

Tracked in full in `docs/future_work.md`. Highlights:

- **GroundFailure episode dedup / rate-limiting** — the real export shows 79,480 events in 14 months on one fleet segment (a chattering-sensor cohort); production needs episode-collapse before alerting.
- **err1051 meter-zero corroboration** on a meter-bearing real fault window (status-spine and transaction-linkage steps are already real-verified — flags 25/27).
- **Temperature features** — a located export (65,994 readings, Body/Outlet sensors) landed post-freeze (flag 27); code and tests are already in place, integration is queued.
- **Autoencoder Layer 2**, natural-language fault query, multi-tenant SaaS, and detectors for the remaining 13 OCPP categories.

10. Team

Name	Responsibility
V. Abhishek Prakash	Layer 1 detectors, prioritizer, replay/detector/UI integration, Docker pipeline
Akhil Prasad	Layer 2 modeling, data audit, taxonomy + normalization, evaluation notebooks
Hrishikesh	Manual testing, documentation review, submission logistics

Provenance: anonymized OCPP telemetry from a production charging-management system.

Charge-box IDs SHA-256-hashed; geo rounded; customer/RFID/IP dropped at export. Raw exports are never committed. MIT licensed.